

Exploring the Impact of K-Space Interpolation and Masking Acceleration on MRI Reconstruction Using VarNet

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Introduction

Magnetic Resonance Imaging (MRI) acquires data in the frequency domain, known as the k-space, which is then transformed into an image via the inverse Fourier transform. To reduce image acquisition time, it is common to undersample the k-space using a masking strategy with an acceleration factor.

The Variational Network (VarNet) is a deep-learning reconstructing method trained to map this undersampled k-space data directly to high-quality images. This work, however, explores a different approach: instead of relying solely on the network to recover missing data, the pre-processing of the images is altered by using interpolation methods on the undersampled k-space to estimate and fill in missing values before passing the data to VarNet. Ding and Zhang were successful in using a deep-learning network to learn fourier interpolation, demonstrating advantages of this pre-processing approach [1].

We aim to investigate how the interpolation methods nearest-neighbor(NN), cubic spline, fourier, and radial basis function (RBF) affect the quality of MRI reconstruction considering mask accelerations factors 4, 6, and 8.

Methods

Nearest-Neighbour Estimates missing values by assigning the value of the closest sampled point.	Cubic Spline Fits smooth cubic polynomials between known points for continuity.
Fourier Interpolation Reconstructs missing data using Fourier transforms and leveraging the periodicity and symmetry of the data for smooth interpolation.	RBF Fills in missing k-space values by using the multiquadric radial basis function, to model the relationship between known data points and then calculates interpolated values based on the distance from the known points.

Table 1 - Interpolation Methods used

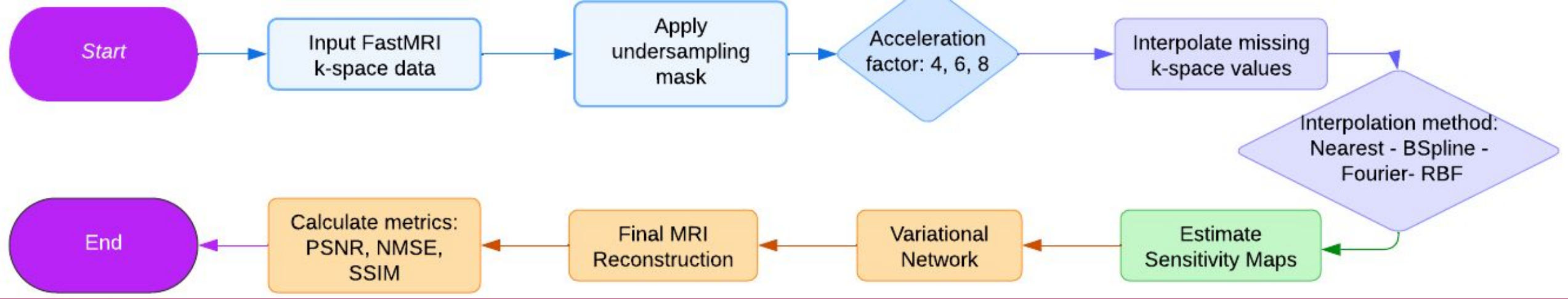


Figure 1 - Flowchart of our model

Input Data: The fastMRI dataset was used for training, evaluation and testing [2]. To ensure reproducibility, the same random mask is used across all experiments by setting a fixed seed. Higher acceleration factors correspond to more aggressive undersampling.

Evaluation metrics

Quantitative Metrics

PSNR Evaluates the reconstruction quality by comparing the peak signal (maximum possible value) to the noise level in the image.	NMSE Quantifies the difference between the ground truth and reconstructed images, normalized by the variance of the ground truth.	SSIM Measures the similarity between the ground truth and the reconstructed images based on luminance, contrast, and structure.
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Qualitative Metrics

A qualitative evaluation is also conducted by visually comparing the ground truth magnitude with the reconstructed magnitude through plotting. This provides insight into the visual fidelity of the reconstruction and the preservation of important image details.

Acceleration factor	Learning rate	Mask type	Epochs
4 (mild)	0.001	random	25
6 (moderate)	0.001	random	40
8 (extreme)	0.001	random	30

Table 2 - Model parameters for training and testing

Results

	NMSE	PNR	SSIM
Basic	0.4126	31.2697	0.7744
NN	0.4230	30.8995	0.7739
Cubic spline	0.4450	29.9540	0.7758
Fourier	0.4034	31.4405	0.7763
RBF	0.3924	31.1788	0.7722

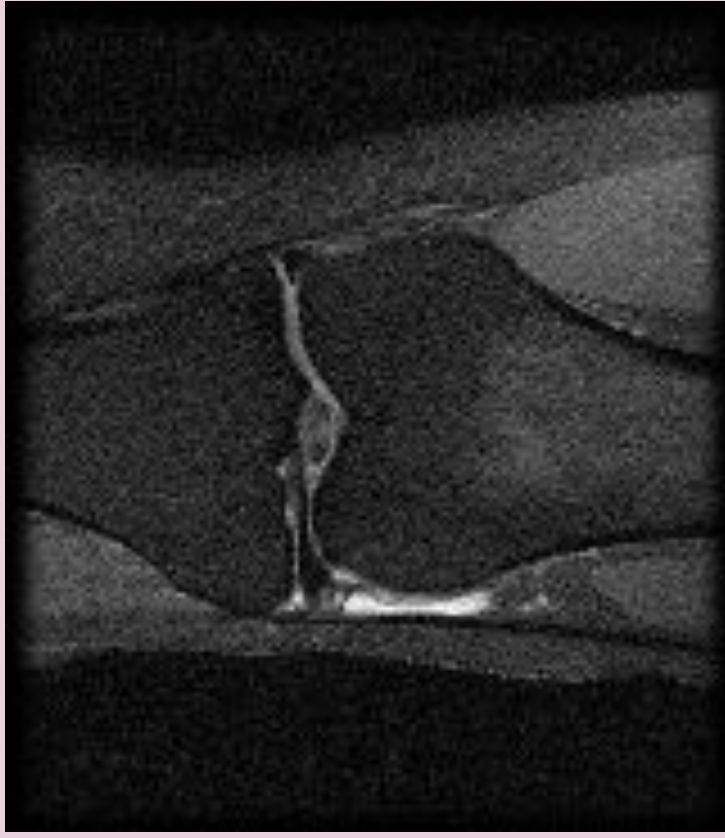
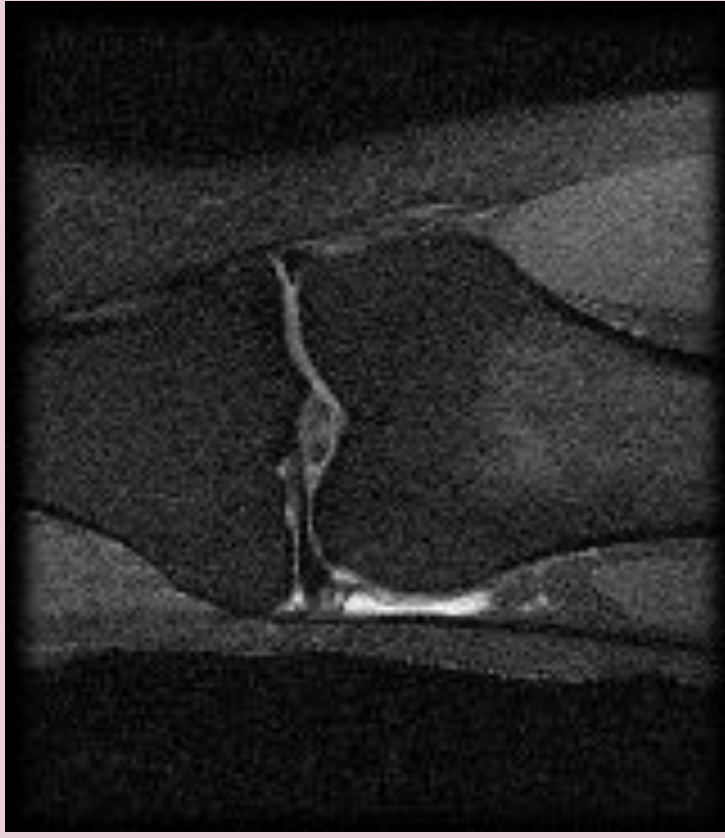
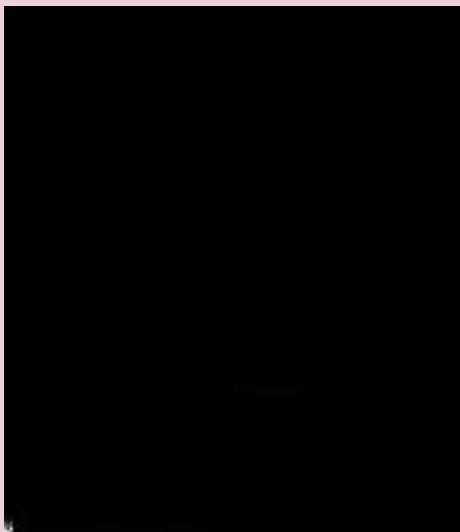
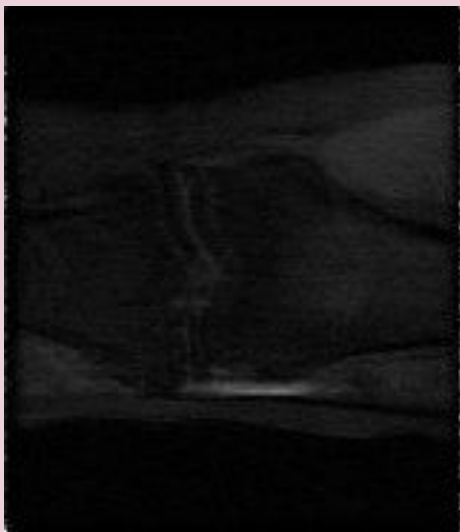
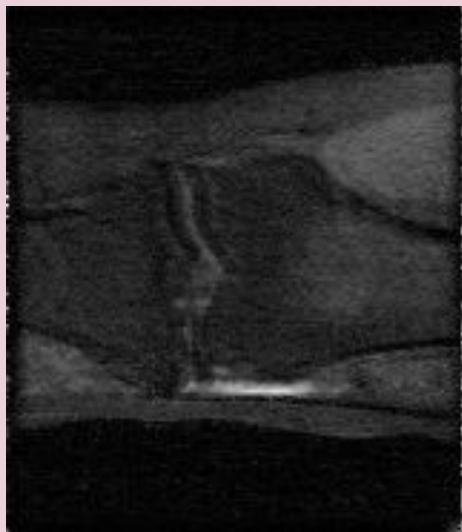
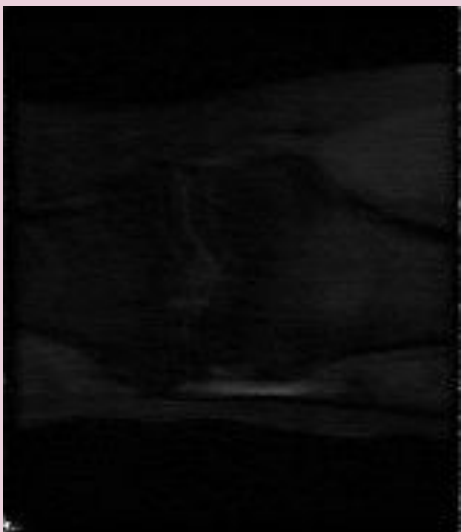
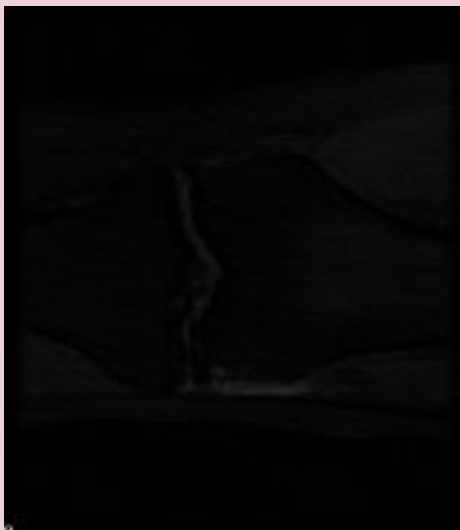
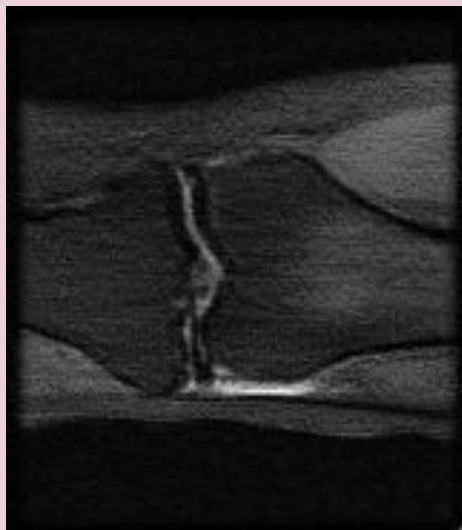
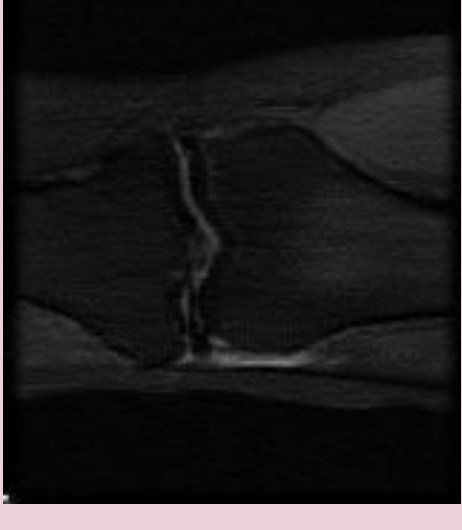
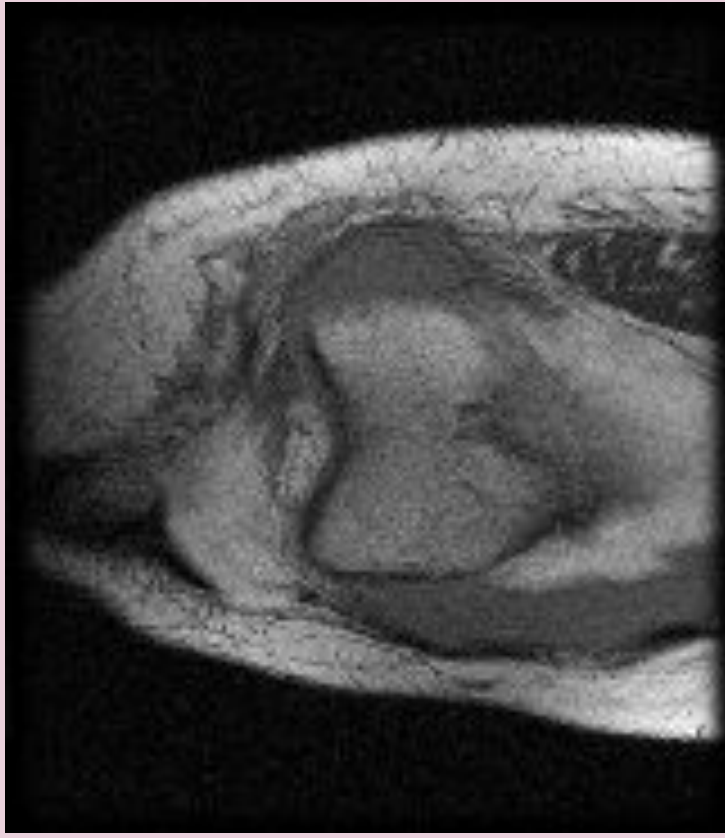
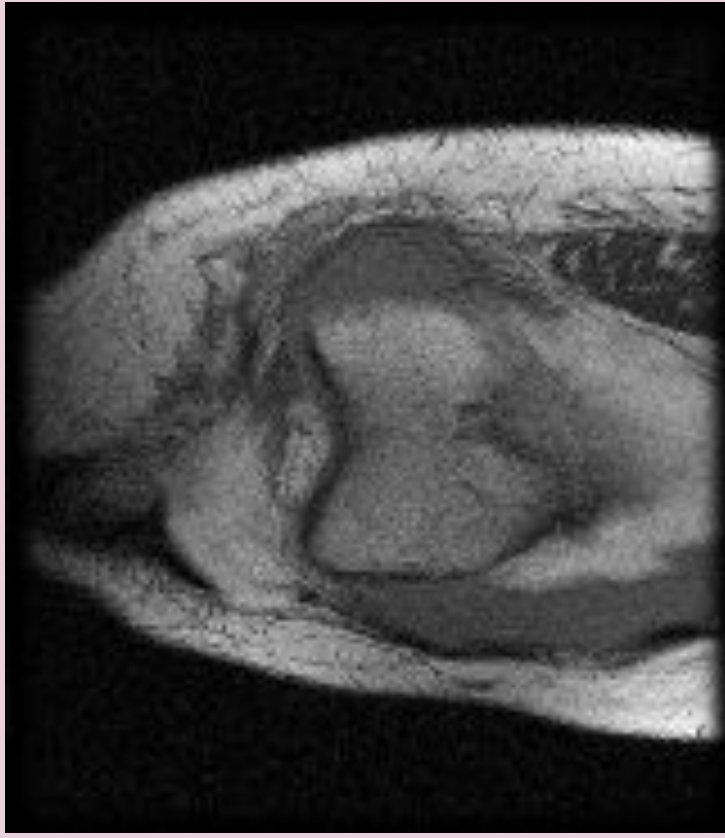
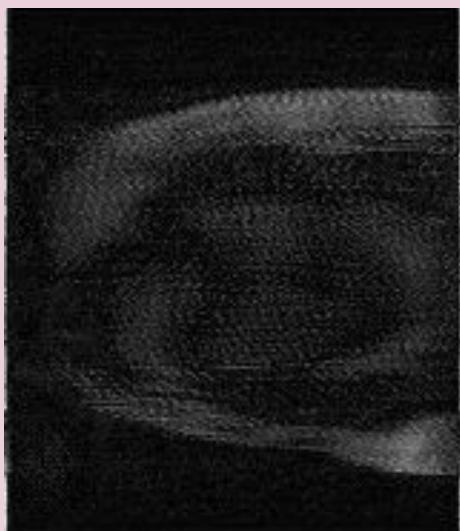
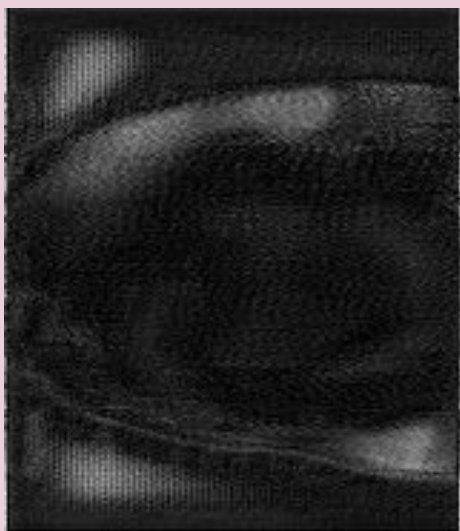
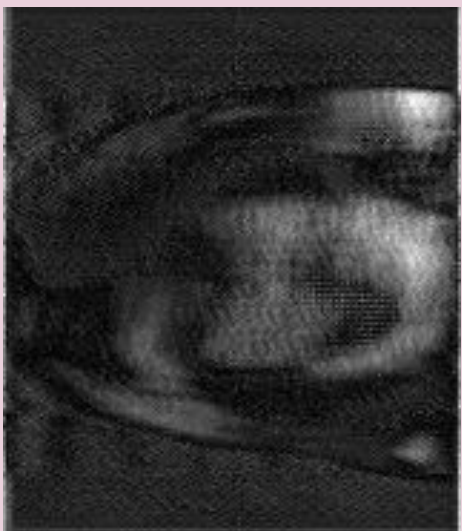
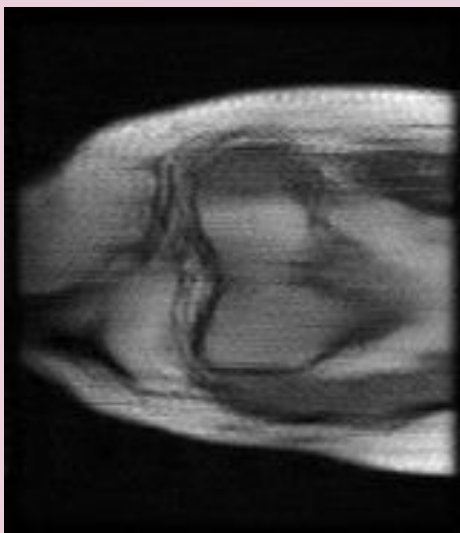
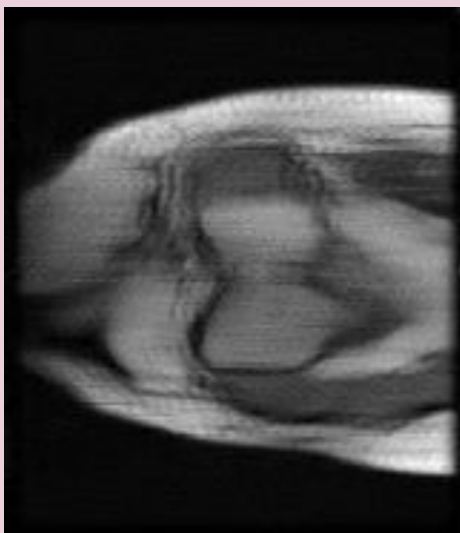
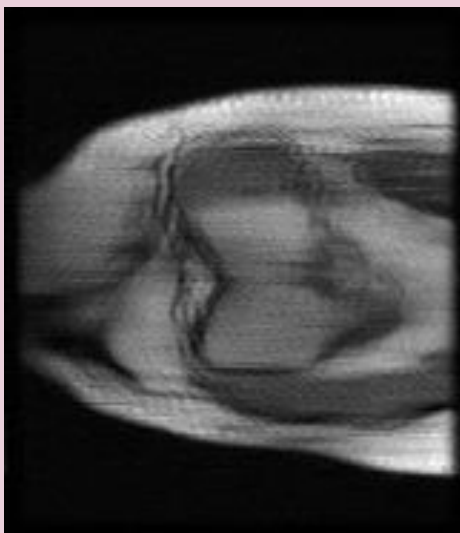
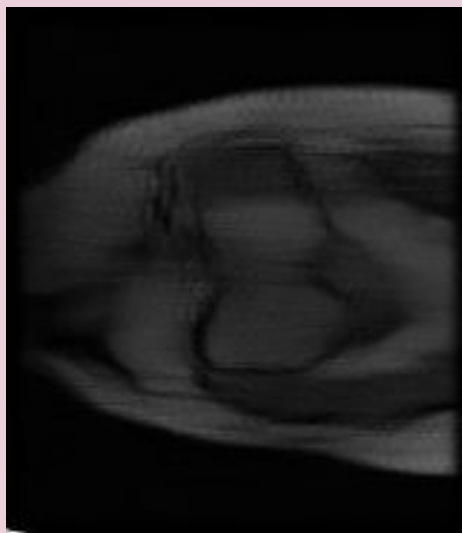
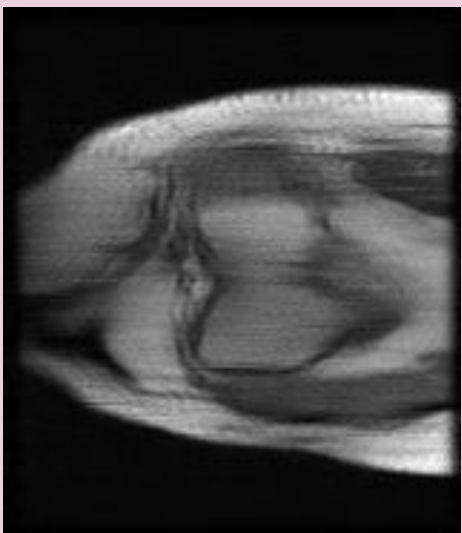
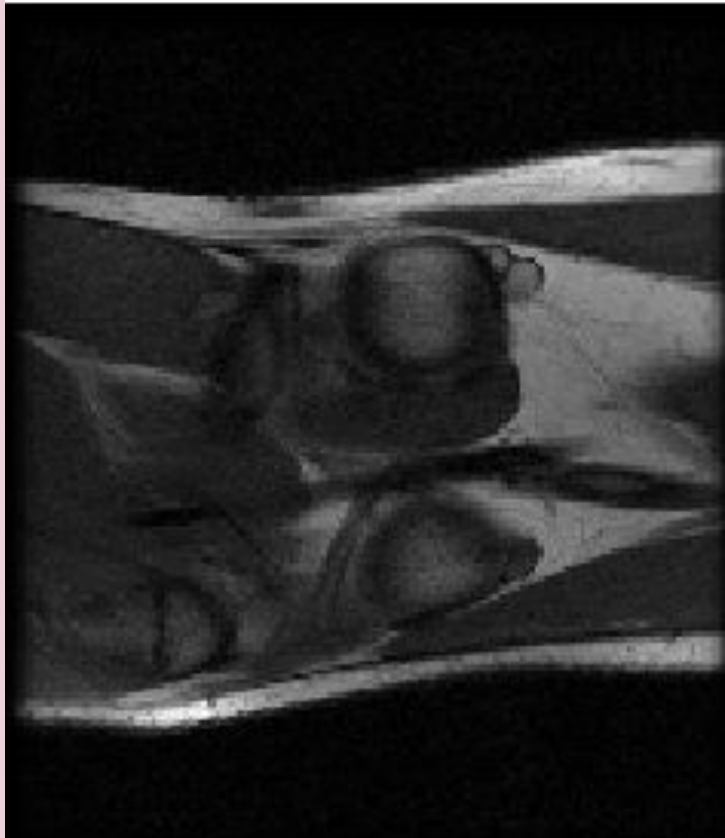
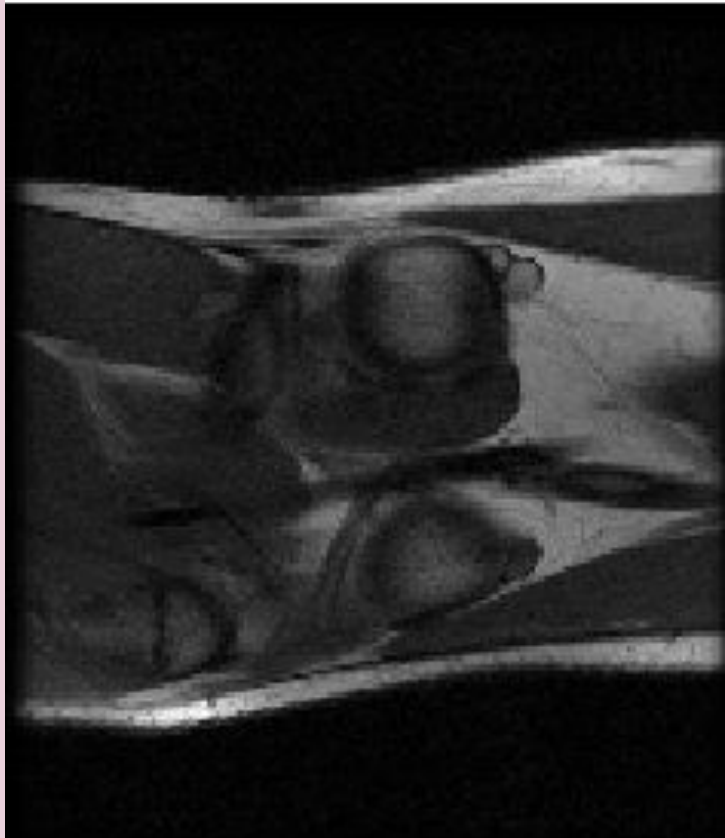
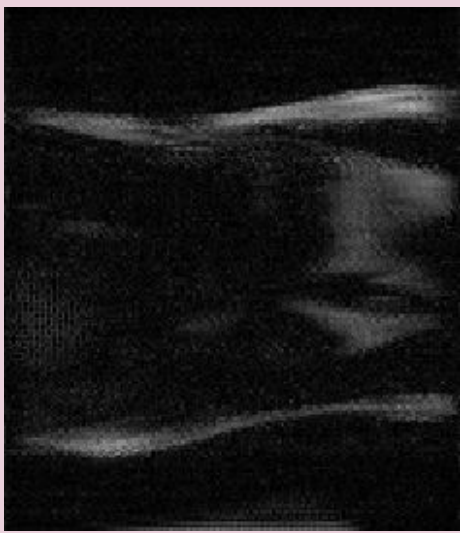
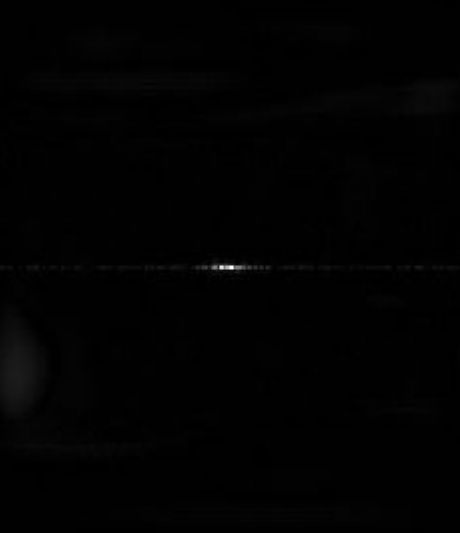
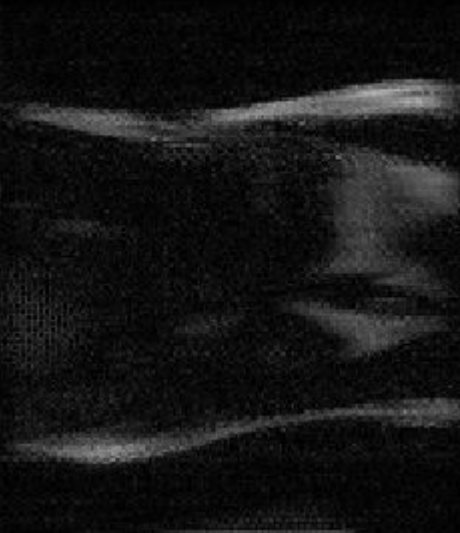
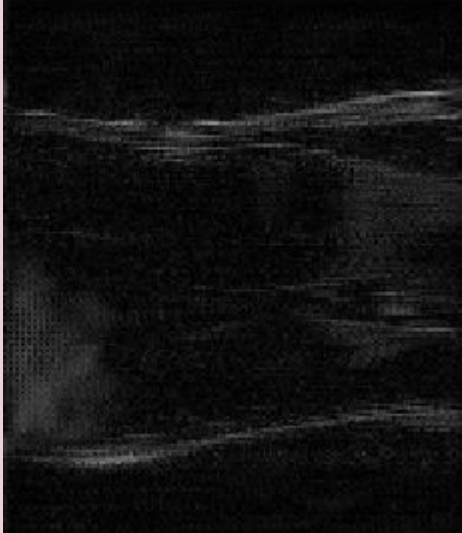
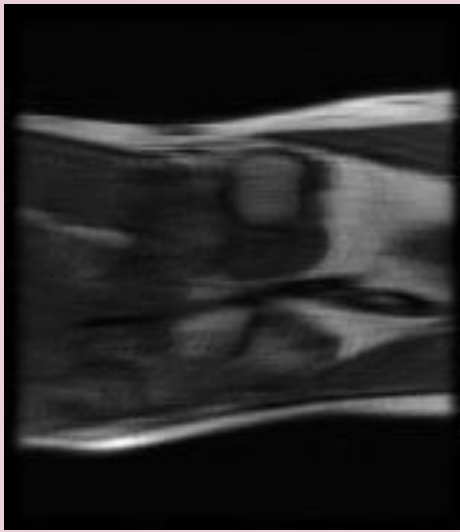
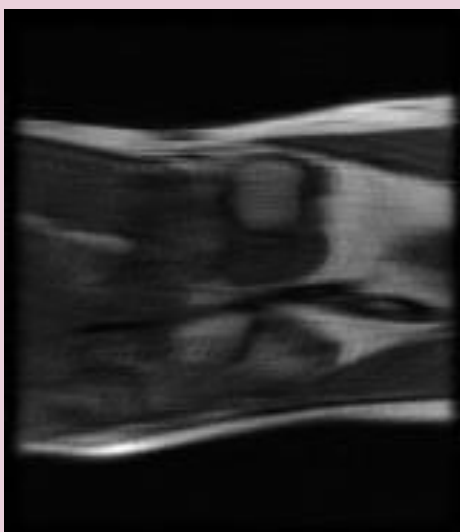
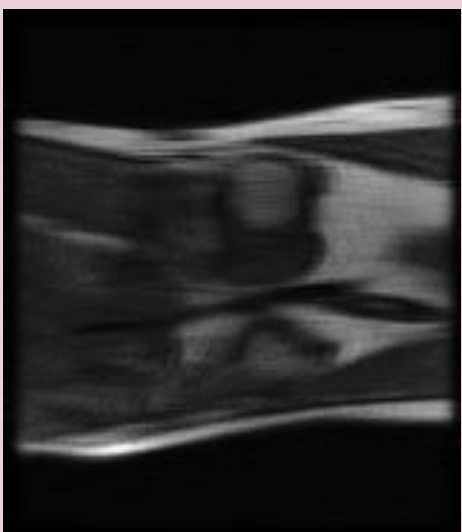
Table 3 - Testing metrics for mask acceleration factor 4

	NMSE	PSNR	SSIM
Basic	0.4240	30.5546	0.7521
NN	0.4215	30.4317	0.7506
Cubic spline	0.4434	29.5820	0.7508
Fourier	0.4116	30.7311	0.7529
RBF	0.4151	30.6524	0.7520

Table 4 - Testing metrics for mask acceleration factor 6

	NMSE	PSNR	SSIM
Basic	0.4214	30.1315	0.7318
NN	5.4495	8.9996	0.7097
Cubic spline	0.4214	30.1315	0.7318
Fourier	0.4089	30.2682	0.7293
RBF	0.4303	29.8644	0.7301

Table 5 - Testing metrics for mask acceleration factor 8

	Acc	GT	Basic	NN	Cubic spline	Fourier	RBF	Epoch
4								First
								Last
6								First
								Last
8								First
								Last

Conclusion

- As acceleration increases, all methods experience increased blurring due to the loss of high-frequency details.
- Fourier interpolation:**Provides the most accurate initial k-space because it preserves the frequency domain's periodicity and symmetry.
- Nearest-neighbour interpolation:** Performs the worst, as expected, because it assigns values based solely on the closest sampled point. If many neighboring values do not offer a sample to learn from, this method fails to predict a value.
- B-spline interpolation:** Introduces many shadows in the initial k-space due to its smooth polynomial approach, which may not handle sharp changes or edges well, resulting in oversmoothing and artifacts that appear as shadows or ghosting in the final image.
- After all epochs, final reconstructions are very similar across all methods, highlighting VarNet’s ability to effectively reconstruct images even with different initial k-spaces. This demonstrates that VarNet's learned regularization and data consistency corrections can overcome the initial interpolation differences, making pre-processing less influential and emphasizing the network's robustness in refining k-space through multiple cascades.

References

[1] Q. Ding and X. Zhang, "MRI Reconstruction by Completing Under-sampled K-space Data with Learnable Fourier Interpolation," in *Medical Image Computing and Computer Assisted Intervention – MICCAI 2022*, L. Wang, Q. Dou, P. T. Fletcher, S. Speidel, and S. Li, Eds., Cham: Springer Nature Switzerland, 2022, pp. 676–685. doi: [10.1007/978-3-031-16446-0_64](https://doi.org/10.1007/978-3-031-16446-0_64).

[2] Knoll et al., "Radiology Artificial Intelligence," *Radiology: Artificial Intelligence*, vol. 2, no. 1, pp. e190007, Jan. 2020, doi: 10.1148/ryai.2020190007 . Available: <https://pubs.rsna.org/doi/10.1148/ryai.2020190007>